Notebook 6 — Chapter 6.2

Optimal Stopping Theory: the Secretary Problem

20 worked examples — problem setup, the $1/e$ threshold, using and stress-testing the rule, and variants — calibrated for Stage 1/Stage 2 speed and rigor

The Setup and the Rule (read this first)

n candidates arrive one at a time in a uniformly random order. After each interview you learn only the candidate's *rank relative to those already seen*. You must accept or reject on the spot (*no recall*), and you win only if you hire the **overall best**.
Rule with cut-off s: reject the first s candidates (pure observation), then hire the first candidate who is better than *everyone seen so far*.

$$\mathbb{P}_n(s) = \frac{s}{n}\sum_{j=s+1}^{n}\frac{1}{j-1} = \frac{s}{n}\sum_{i=s}^{n-1}\frac{1}{i} \qquad (s \geq 1)$$

As $n \to \infty$ with $x = s/n$: $\mathbb{P} \to -x\ln x$, maximised at $x = \frac{1}{e}$ with value $\frac{1}{e} \approx 0.368$.

A. Setup, Small Cases and the Key Lemma

Example 1 — Three Candidates, Brute Force

Three candidates have qualities $1 < 2 < 3$ (3 is best) and arrive in a random order. Use the rule "reject the first, then hire the first candidate better than everyone seen". Find $\mathbb{P}$(hire the best).
Solution. List all $3! = 6$ orders:

order	outcome	win?
$(1,2,3)$	2 beats 1: hire 2	×
$(1,3,2)$	3 beats 1: hire 3	✓
$(2,1,3)$	1 no; 3 beats 2: hire 3	✓
$(2,3,1)$	3 beats 2: hire 3	✓
$(3,1,2)$	nobody beats 3: no hire	×
$(3,2,1)$	nobody beats 3: no hire	×

$$\mathbb{P}(\text{win}) = \frac{3}{6} = \boxed{\tfrac{1}{2}}$$

Hiring the first candidate blindly wins only $\frac{1}{3}$.
⋆ *Analogy:* the first candidate is a *calibration interview*. You do not intend to hire them — you are measuring how good "good" looks in this pool.

Example 2 — Four Candidates: Choosing the Cut-off

For $n = 4$ compute $\mathbb{P}_4(s)$ for each cut-off and pick the best.
Solution. Apply $\mathbb{P}_n(s) = \frac{s}{n}\sum_{i=s}^{n-1}\frac{1}{i}$:

$$s=1:\ \tfrac{1}{4}\left(1+\tfrac{1}{2}+\tfrac{1}{3}\right) = \tfrac{1}{4}\cdot\tfrac{11}{6} = \tfrac{11}{24} \approx 0.458$$

$$s=2:\ \tfrac{2}{4}\left(\tfrac{1}{2}+\tfrac{1}{3}\right) = \tfrac{1}{2}\cdot\tfrac{5}{6} = \tfrac{5}{12} \approx 0.417$$

$$s=3:\ \tfrac{3}{4}\cdot\tfrac{1}{3} = \tfrac{1}{4}, \qquad s=0 \text{ (hire the first)}:\ \tfrac{1}{4}$$

The best is $\boxed{s=1,\ \mathbb{P}=\frac{11}{24}}$.
▶ *Speed trick:* the success probability is *unimodal* in s: it rises while you are still learning, then falls once you throw away too many candidates. Stop computing as soon as it starts to drop.

Example 3 — Deriving the Success Formula ($n = 5$)

Derive $\mathbb{P}_n(s)$ and evaluate it for $n = 5$, $s = 1, 2, 3, 4$.
Solution. Condition on the position j of the overall best candidate; $\mathbb{P}$(best at j) $= \frac{1}{n}$. We win iff $j > s$ *and* the best of the first $j-1$ candidates lies among the first s (otherwise a record in positions $s+1, \ldots, j-1$ would be hired early). By symmetry that happens with probability $\frac{s}{j-1}$:

$$\mathbb{P}_n(s) = \sum_{j=s+1}^{n}\frac{1}{n}\cdot\frac{s}{j-1} = \frac{s}{n}\sum_{i=s}^{n-1}\frac{1}{i}$$

$$n=5:\ s=1:\ \tfrac{1}{5}\cdot\tfrac{25}{12} = \tfrac{5}{12}\ (0.417), \qquad s=2:\ \tfrac{2}{5}\cdot\tfrac{13}{12} = \boxed{\tfrac{13}{30}}\ (0.433)$$

$$s=3:\ \tfrac{3}{5}\cdot\tfrac{7}{12} = \tfrac{7}{20}\ (0.35), \qquad s=4:\ \tfrac{4}{5}\cdot\tfrac{1}{4} = \tfrac{1}{5}\ (0.2)$$

Best: $s = 2$.
⋆ *Analogy:* the best candidate is a *lost needle*. You find it iff the strongest rival before it was already filed away in the observation pile — otherwise you will have hired that rival.

Example 4 — The Optimal Cut-off Table

Find the optimal s and $\mathbb{P}_n(s)$ for $n = 1, \ldots, 10$.
Solution. Maximise $\mathbb{P}_n(s)$ over s for each n:

n	1	2	3	4	5	6	7	8	9	10
$s^\star$	0	0	1	1	2	2	2	3	3	3
$\mathbb{P}$	1	.500	.500	.458	.433	.428	.414	.410	.406	.399

(For $n = 2$, $s = 0$ and $s = 1$ tie at $\frac{1}{2}$.) The cut-off grows like n/e while the success probability slides down toward 0.368.
▶ *Speed trick:* for $n \geq 8$ just use $s = \text{round}(n/e)$; the loss from the approximation is below 0.001.

Example 5 — The Record Lemma

Show that candidate j is a "record" (best so far) with probability $\frac{1}{j}$, independently across j. For $n = 10$ find $\mathbb{E}$[#records] and $\mathbb{P}$(no record among positions $6, \ldots, 10$).
Solution. Among the first j candidates, the best is equally likely to be in any of the j positions, so $\mathbb{P}(I_j = 1) = \frac{1}{j}$. The relative rank of candidate j among the first j is uniform on $\{1, \ldots, j\}$ and independent of the earlier relative ranks, which gives independence.

$$\mathbb{E}[\#\text{records}] = \sum_{j=1}^{10}\frac{1}{j} = H_{10} = \tfrac{7381}{2520} \approx \boxed{2.93}$$

$$\mathbb{P}(\text{no record in } 6..10) = \prod_{j=6}^{10}\left(1-\tfrac{1}{j}\right) = \tfrac{5}{6}\cdot\tfrac{6}{7}\cdot\tfrac{7}{8}\cdot\tfrac{8}{9}\cdot\tfrac{9}{10} = \boxed{\tfrac{1}{2}}$$

In general, no record in $s+1, \ldots, n$ has probability $\frac{s}{n}$.
⋆ *Analogy:* the telescoping product is a *conveyor belt of fresh challengers*: the chance that nobody on the last $n - s$ slots beats the leader so far is simply the fraction of the belt that was already behind you.

B. Deriving the $1/e$ Rule

Example 6 — The $-x\ln x$ Limit

Let $n \to \infty$ with $s/n \to x$. Show $\mathbb{P} \to -x\ln x$ and maximise.
Solution. The harmonic sum is a Riemann sum for $\int dt/t$:

$$\sum_{i=s}^{n-1}\frac{1}{i} \longrightarrow \ln\frac{n}{s} = -\ln x$$

$$\mathbb{P}_n(s) = \frac{s}{n}\sum_{i=s}^{n-1}\frac{1}{i} \longrightarrow \phi(x) = -x\ln x$$

Differentiate:

$$\phi'(x) = -\ln x - 1 = 0 \ \Rightarrow\ x^\star = \tfrac{1}{e}, \qquad \phi''(x) = -\tfrac{1}{x} < 0$$

$$\phi(x^\star) = -\tfrac{1}{e}\ln\tfrac{1}{e} = \boxed{\tfrac{1}{e} \approx 0.368}$$

⋆ *Analogy:* observation is an investment: skip too little and you cannot tell a star from a pretender; skip too much and the star may already have walked past. The break-even fraction is 36.8%.

Example 7 — How Fast Does It Converge?

Compute the exact optimal s and $\mathbb{P}$ for $n = 20, 50, 100, 1000$ and compare with n/e and $1/e$.
Solution. Evaluate $\frac{s}{n}\sum_{i=s}^{n-1}\frac{1}{i}$ and maximise over s:

n	20	50	100	1000
n/e	7.36	18.39	36.79	367.88
$s^\star$	7	18	37	368
$\mathbb{P}$	.3842	.3743	.3710	.3682

The limit is $\frac{1}{e} = 0.3679$. At $n = 100$ the exact optimum is 0.3710 — the continuous formula is already accurate to 3×10^{-3}.
▶ *Speed trick:* the optimal cut-off is always the integer nearest n/e (here exactly so), and the finite-n value approaches $1/e$ *from above*.

Example 8 — A Flat Top: Sensitivity to the Cut-off

Evaluate $\phi(x) = -x\ln x$ at $x = 0.2, 0.25, 0.3, 0.4, \frac{1}{2}, 0.6$ and find the range of x giving at least 0.35 and at least 0.30.
Solution.

x	.2	.25	.3	.4	.5	.6
ϕ	.322	.347	.361	.367	.347	.306

Note $\phi(\frac{1}{4}) = \frac{1}{4}\ln 4 = \frac{\ln 2}{2} = \phi(\frac{1}{2})$ exactly. Solving $\phi(x) = 0.35$ numerically:

$$\phi(x) \geq 0.35 \iff x \in [0.259,\ 0.488]$$

$$\phi(x) \geq 0.30 \iff x \in [0.168,\ 0.613]$$

⋆ *Analogy:* the peak is a *broad plateau*, not a needle. Anywhere from a quarter to a half of the pool as exploration costs you at most 0.02 from the optimum.

Example 9 — Win, Wrong Hire, or No Hire?

With $n = 100$ and $s = 37$, find $\mathbb{P}$(best hired), $\mathbb{P}$(nobody hired) and $\mathbb{P}$(wrong person hired). Give the $n \to \infty$ values.
Solution. The rule hires nobody iff there is no record after position s, i.e. iff the best candidate is in the first s:

$$\mathbb{P}(\text{no hire}) = \frac{s}{n} = 0.370$$

$$\mathbb{P}(\text{win}) = \mathbb{P}_{100}(37) = 0.3710$$

$$\mathbb{P}(\text{wrong hire}) = 1 - 0.3710 - 0.370 = \boxed{0.2590}$$

As $n \to \infty$ with $x = \frac{1}{e}$: win $= \frac{1}{e}$, no hire $= \frac{1}{e}$, wrong hire $= 1 - \frac{2}{e} \approx 0.264$.
▶ *Speed trick:* at $s = n/e$ the two "extreme" outcomes, winning and hiring nobody, are equally likely. The *only* way to hire somebody mediocre is a decoy record that sits in the middle.

Example 10 — The Dynamic-Programming Threshold

Show that the optimal rule stops at a record iff $\sum_{i=j}^{n-1} \frac{1}{i} \leq 1$, and find the first such j for $n = 10$, $n = 100$.
Solution. A record at time j is the overall best with probability $\frac{j}{n}$ (the best of n is equally likely in any of the first j positions' "best so far"). Stopping wins with $\frac{j}{n}$; continuing with the threshold rule from j wins with $\frac{j}{n}\sum_{i=j}^{n-1}\frac{1}{i}$. Stop iff

$$\frac{j}{n} \geq \frac{j}{n}\sum_{i=j}^{n-1}\frac{1}{i} \iff \sum_{i=j}^{n-1}\frac{1}{i} \leq 1$$

The sum decreases in j, so the rule is a threshold at the first r with $\sum_{i=r}^{n-1}\frac{1}{i} \leq 1$.
$n = 10$: $\sum_{i=4}^{9}\frac{1}{i} = 0.9956 \leq 1$ but $\sum_{i=3}^{9}\frac{1}{i} = 1.329 > 1$, so $r = 4$ (skip 3).
$n = 100$: $\sum_{i=38}^{99}\frac{1}{i} = 0.9758 \leq 1$ but $\sum_{i=37}^{99}\frac{1}{i} = 1.0028 > 1$, so $r = 38$ (skip $\boxed{37}$).
Large n: $\ln\frac{n}{r} = 1 \Rightarrow r = \frac{n}{e}$.
⋆ *Analogy:* the "sum of odds ≤ 1" test: each remaining slot i is a chance of $\frac{1}{i}$ that a record appears; once the total expected future opportunities drop to one, grabbing a record now beats waiting.

C. Using and Stress-Testing the Rule

Example 11 — Stop or Keep Waiting? ($n = 20$)

With $n = 20$ and a record appearing at time j, compare $\mathbb{P}$(win by stopping) with $\mathbb{P}$(win by continuing optimally) for $j = 5, \ldots, 11$.
Solution. Stop: $\frac{j}{20}$. Continue: $\frac{j}{20}\sum_{i=j}^{19}\frac{1}{i}$.

j	5	6	7	8	9	10	11
stop	.250	.300	.350	.400	.450	.500	.550
continue	.366	.379	.384	.382	.373	.359	.340

Continuing wins for $j \leq 7$; stopping wins from $j = 8$ on. So skip the first $\boxed{7}$ and hire the first record from $j = 8$, matching Example 7.
▶ *Speed trick:* compare the two rows and find the *crossing*. Stopping probability rises linearly while the continuation value is a gentle hump.

Example 12 — How Many Interviews Will You Run?

For $n = 100$, $s = 37$ find $\mathbb{E}$[#interviews].
Solution. The rule stops at position $j > s$ iff j is a record and the best of the first $j-1$ is in the first s:

$$\mathbb{P}(\text{stop at } j) = \frac{1}{j}\cdot\frac{s}{j-1}$$

Stopping times sum to $n\cdot\mathbb{P}(\text{win})$:

$$\sum_{j=s+1}^{n} j\cdot\frac{s}{j(j-1)} = s\sum_{i=s}^{n-1}\frac{1}{i} = n\,\mathbb{P}_n(s) = 37.10$$

If nobody qualifies (probability $\frac{s}{n}$) you interview all n, contributing $n\cdot\frac{s}{n} = s$:

$$\mathbb{E}[T] = n\,\mathbb{P}_n(s) + s = 37.10 + 37 = \boxed{74.10}$$

(simulation: 74.0). Asymptotically $\frac{2n}{e} \approx 0.736\,n$.
⋆ *Analogy:* you will on average see about three-quarters of the pool — the observation phase is not free, it is the price of a 37% shot.

Example 13 — What If n Was Wrong?

You planned for $n = 100$ and fixed the cut-off at 37 candidates. Find the success probability if the pool turns out to have $n = 50, 60, 80, 150, 200, 400$.
Solution. With $s = 37$ fixed, $\mathbb{P}_n(37) = \frac{37}{n}\sum_{i=37}^{n-1}\frac{1}{i}$:

n	50	60	80	150	200	400
$\mathbb{P}_n(37)$	.225	.301	.360	.348	.314	.221
skip n/e	.374	.373	.372	.370	.370	.369

A fixed *count* is fragile, a fixed *fraction* is robust: using n/e of whatever pool arrives always gives ≈ 0.37.
▶ *Speed trick:* if n is uncertain, *observe for a fraction of elapsed time*, not a fixed number of candidates.

Example 14 — Given That You Hired Someone

For $n = 100$, $s = 37$, find $\mathbb{P}$(best | somebody hired) and its limit.
Solution. $\mathbb{P}(\text{hire}) = 1 - \frac{s}{n} = 0.63$:

$$\mathbb{P}(\text{best} \mid \text{hired}) = \frac{0.3710}{0.63} = \boxed{0.589}$$

As $n \to \infty$:

$$\frac{1/e}{1-1/e} = \frac{1}{e-1} \approx 0.582$$

⋆ *Analogy:* the rule is a *pass-or-fail exam*. About a third of the time nobody passes; in the two-thirds of runs where somebody does, the pick is the true best nearly 60% of the time.

Example 15 — Counting Check for $n = 5$, $s = 2$

Verify $\mathbb{P}_5(2) = \frac{13}{30}$ by counting permutations of $\{1, \ldots, 5\}$.
Solution. There are $5! = 120$ orders; each position of the best (candidate 5) has $4! = 24$ orders. The rule wins iff the best of the first $j-1$ is among the first 2, probability $\frac{2}{j-1}$:

$$j = 1, 2:\ 0 \text{ (best is in the observation set)}$$

$$j = 3:\ 24\cdot\tfrac{2}{2} = 24, \quad j = 4:\ 24\cdot\tfrac{2}{3} = 16, \quad j = 5:\ 24\cdot\tfrac{2}{4} = 12$$

$$\text{wins} = 24 + 16 + 12 = 52, \qquad \frac{52}{120} = \boxed{\tfrac{13}{30}}$$

(a full computer enumeration of all 120 orders gives 52 wins).
▶ *Speed trick:* for tiny n, count wins by position of the best; the factor $\frac{s}{j-1}$ is the whole story.

D. Variants and Capstones

Example 16 — Fifty Sequential Quotes

A trader receives 50 price quotes one by one from different dealers, sees each only against the earlier ones, must accept or reject at once and cannot recall. How many should she let pass, and what is her chance of taking the best quote?
Solution. This is the secretary problem with $n = 50$. $n/e = 18.39$;

checking the sums:

$$\sum_{i=19}^{49} \tfrac{1}{i} = 0.9841 \ \ (\le 1), \qquad \sum_{i=18}^{49} \tfrac{1}{i} = 1.0397 \ \ (> 1)$$

so skip $s = 18$ and take the first quote better than all earlier ones:

$$\mathbb{P} = \frac{18}{50} \sum_{i=18}^{49} \frac{1}{i} = \boxed{0.3743}$$

⋆ *Analogy:* the first 18 quotes are *market reconnaissance*. They set the "best price so far" benchmark that all later quotes must beat.

Example 17 — Bruss's Odds Theorem

Independent events $I_1, \dots, I_8$ occur with probabilities $p = (\frac{1}{10}, \frac{1}{5}, \frac{3}{20}, \frac{3}{10}, \frac{1}{4}, \frac{1}{5}, \frac{3}{10}, \frac{2}{5})$. You observe them in order and want to stop on the *last* success. Find the optimal rule and its success probability.

Solution. Odds $r_k = \frac{p_k}{1-p_k}$:

$$r = (\tfrac{1}{9}, \tfrac{1}{4}, \tfrac{3}{17}, \tfrac{3}{7}, \tfrac{1}{3}, \tfrac{1}{4}, \tfrac{3}{7}, \tfrac{2}{3})$$

Add odds from the end until the sum reaches 1:

$$r_8 = \tfrac{2}{3} < 1, \qquad r_8 + r_7 = \tfrac{2}{3} + \tfrac{3}{7} = \tfrac{23}{21} \ge 1 \ \Rightarrow \ s = 7$$

Stop at the first success from index 7 on:

$$\mathbb{P}(\text{win}) = \prod_{k=7}^{8} (1 - p_k) \cdot \sum_{k=7}^{8} r_k = \tfrac{7}{10} \cdot \tfrac{3}{5} \cdot \tfrac{23}{21} = \boxed{\tfrac{23}{50} = 0.46}$$

(a backward dynamic program and a simulation both give 0.46).

▶ *Speed trick: sum of odds from the back until it hits* 1 — no dynamic programming needed.

Example 18 — The Secretary Problem Is an Odds Problem

Rederive the $n = 10$ cut-off and win probability from the odds theorem, using record indicators I_k with $p_k = \frac{1}{k}$.

Solution. The record indicators are independent (Example 5) and the best candidate is the *last* record. Odds: $r_k = \frac{1/k}{1-1/k} = \frac{1}{k-1}$. Sum from the back:

$$\tfrac{1}{9}, \ \tfrac{1}{9}+\tfrac{1}{8}, \ \cdots : \quad 0.111, \ 0.236, \ 0.379, \ 0.546, \ 0.746, \ 0.996, \ \underbrace{1.329}_{k=4}$$

The sum first reaches 1 at $k = 4$: skip $s = 3$ (Example 4). Then

$$\mathbb{P} = \prod_{k=4}^{10} \left(1 - \tfrac{1}{k}\right) \cdot 1.329 = \tfrac{3}{10} \cdot 1.329 = \boxed{0.3987}$$

exactly the entry for $n = 10$ in Example 4.

⋆ *Analogy:* "hire the last record" and "stop on the last success" are the same game. The odds theorem is the secretary problem with the dials released — the $\frac{1}{k}$ can be *any* sequence of independent probabilities.

Example 19 — What If You See the Actual Scores?

Candidates have iid Uniform$(0, 1)$ scores and you *see the score*, still wanting the overall maximum. Solve $n = 2$, and compare with the rank-only game.

Solution. Candidate 1 has score x. Accepting wins iff candidate 2 is lower: probability x. Rejecting and taking candidate 2 (forced) wins iff $y > x$: probability $1 - x$. Accept iff $x > 1 - x$, i.e. $x > \frac{1}{2}$:

$$\mathbb{P}(\text{win}) = \int_{1/2}^{1} x \, dx + \int_{0}^{1/2} (1 - x) \, dx = \tfrac{3}{8} + \tfrac{3}{8} = \boxed{\tfrac{3}{4}}$$

Rank-only ($n = 2$) gives $\frac{1}{2}$. With m draws left, accept a record x iff x^m beats the continuation value; the same recursion gives (numerically)

$$n = 3: \ 0.684, \qquad n = 10: \ 0.609, \qquad n \to \infty: \ 0.580$$

⋆ *Analogy:* seeing the actual numbers is like reading the thermometer instead of only knowing "warmer or colder". It lifts the long-run success from 0.368 to 0.580.

Example 20 — Capstone: Twenty Interviews

You will interview $n = 20$ candidates. Find the optimal cut-off, $\mathbb{P}$(best), $\mathbb{P}$(no hire), $\mathbb{P}$(wrong hire), $\mathbb{E}$[#interviews], and decide whether to hire a record at $j = 12$.

Solution. Cut-off: $\sum_{i=8}^{19} \frac{1}{i} = 0.955 \le 1 < 1.098 = \sum_{i=7}^{19} \frac{1}{i}$, so $r = 8$: skip $\boxed{s = 7}$.

Success:

$$\mathbb{P} = \frac{7}{20} \sum_{i=7}^{19} \frac{1}{i} = 0.35 \cdot 1.0977 = \boxed{0.3842}$$

No hire: $\frac{s}{n} = \frac{7}{20} = 0.35$. *Wrong hire:* $1 - 0.3842 - 0.35 = 0.2658$.

Interviews: $\mathbb{E}[T] = n\,\mathbb{P} + s = 20(0.3842) + 7 = \boxed{14.68}$ (simulation: 14.69).

Record at $j = 12$: $\mathbb{P}(\text{best}) = \frac{12}{20} = 0.6$; continuing wins only $0.6 \sum_{i=12}^{19} \frac{1}{i} = 0.317$. **Hire.**

▶ *Full-circle intuition:* one recursion — "compare the chance that this record is the best ($\frac{j}{n}$) with the chance of beating it later" — produces the cut-off, the success probability, the $1/e$ limit, the stopping-time law, and every variant in this chapter.